



**Verified Carbon
Standard**

UZUNDERE I 63.0 MW HYDROELECTRIC POWER PLANT PROJECT, TURKEY

Document Prepared by Sekans Danışmanlık

Project Title	Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey
Version	7
Report ID	UZNMONITORING2
Date of Issue	11/02/2022
Project ID	964
Monitoring Period	02 December 2011 – 30 November 2017
Prepared By	SEKANS DANIŞMANLIK
Contact	Konaklar Mah.34330, Yeni Levent-İstanbul E-mail: sila@sekansdanismanlik.com Phone: +90 532 438 30 29

CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Implementation Status of the Project.....	4
1.2	Sectoral Scope and Project Type.....	6
1.3	Project Proponent	7
1.4	Other Entities Involved in the Project	7
1.5	Project Start Date	7
1.6	Project Crediting Period	7
1.7	Project Location.....	8
1.8	Title and Reference of Methodology	8
1.9	Participation under other GHG Programs.....	9
1.10	Other Forms of Credit.....	9
1.11	Sustainable Development	9
2	SAFEGUARDS.....	10
2.1	No Net Harm	10
2.2	Local Stakeholder Consultation	11
2.3	AFOLU-Specific Safeguards	11
3	IMPLEMENTATION STATUS	11
3.1	Implementation Status of the Project Activity	12
3.2	Deviations	12
3.2.1	Methodology Deviations	12
3.2.2	Project Description Deviations.....	12
3.3	Grouped Projects	12
4	DATA AND PARAMETERS.....	12
4.1	Data and Parameters Available at Validation.....	12
4.2	Data and Parameters Monitored.....	13
4.3	Monitoring Plan.....	17
5	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	19

5.1	Baseline Emissions	19
5.2	Project Emissions	22
5.3	Leakage.....	23
5.4	Net GHG Emission Reductions and Removals	23

1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey, which is developed by KARADENİZ HES ELEKTRİK A.Ş. (hereafter referred to as the “project owner”) is a run-of-river hydropower project located in Rize province, Turkey. The project is on the Uzundere creek. The altitude is 817.59 m at powerhouse location.

The Project has been implemented and operated by KARADENİZ HES ELEKTRİK A.Ş.. The project aims to generate electricity from hydropower and feed it to the national electricity grid.

Total installed capacity of the project is 63 MWe, consisting of 2 turbines, with an estimated power supply to the grid of 156,205 MWh per annum. The project is in implementation since May 27th, 2010, when one of the two vertical shaft Pelton turbines in the plant commences electricity generation. The starting date of the project activity for the other turbine has been November 7th, 2010. The estimated amount of GHG emission reduction is 87,318 tonnes CO₂e per year in the registered PDD. During its operation during this monitoring period, the actualized net electricity generation is 811,962.21 MWh. By displacing equivalent amount of electricity consumed from the national grid, which is mostly comprised of fossil fired power plants, the actual emission reduction has been calculated as 453,882 tonnes CO₂ for this monitoring period.

The implementation of the project consists of two diversion weirs, where water from the river is diverted into conveyance pipes; a conveyance tunnel with a length of 11.324 m and powerhouse with two Pelton type vertical turbines. Each turbine has a capacity of 31.534 MWm / 31.076 MWe.

Total installed capacity in the powerhouse is 63.068 MWm / 62.152 MWe). There are two generators attached to the facility. Generators have power factor of 0,9 d/d, a frequency of 50 Hz and an output of 34.5 MVA.

The construction of the project started in 01/04/2010 and the project was operational on 27/05/2010. There is no dam and there hasn't been any increase in the reservoir area during this monitoring period.

The project has produced positive environmental benefits as displacing the electricity generated by fossil fuel fired power plants by utilising the renewable resources so as to avoid environmental pollution and GHG emissions.

Please see the vintage-based comparison of expected and achieved emission reductions for this monitoring period:

Table 1- Expected vs achieved emission reductions

Vintage	Period	Total Days	Amount achieved during this monitoring period (tCO ₂ e)	Amount estimated ex ante (tCO ₂ e)	Difference (tCO ₂ e)	Difference (%)
2011	02.12.2011 - 31.12.2011	30	2,473	7,177	-4,704	-65.5%
2012	01.01.2012 - 31.12.2012	365	65,959	87,318	-21,359	-24.5%
2013	01.01.2013 - 31.12.2013	365	66,784	87,318	-20,534	-23.5%
2014	01.01.2014 - 31.12.2014	365	75,036	87,318	-12,282	-14.1%
2015	01.01.2015 - 31.12.2015	365	71,657	87,318	-15,661	-17.9%
2016	01.01.2016 - 31.12.2016	365	97,304	87,318	9,986	11.4%
2017	01.01.2017 - 30.11.2017	334	74,669	79,902	-5,233	-6.5%
Total		2,189	453,882	523,669	-69,787	-13.3%

As it may be seen through Table - 1, a high increase of generation and amount of emission reduction was realized in 2016. The reason for that is the significant increase in rainfall in the region of the Project Activity for the same year¹. Additionally, surface flow conditions of rainfall and the fall time differences have also an impact on the amount of water in rivers.

Table 2 - Monitoring timeline of the Project Activity

Date (Period)	Activity
01/04/2010	Issuance of EIA Certificate
01/04/2010	Construction Start Date
27/05/2010	Commissioning of the Project
27/05/2010 - 01/12/2011	1 st Monitoring Period
02/12/2011 - 30/11/2017	2 nd Monitoring Period

¹ <https://mgm.gov.tr/veridegerlendirme/yillik-toplam-yagis-verileri.aspx> (Please see the graph "Karadeniz Bölgesi Yıllık Alansal Yağışları")

Please see the technical details of the turbines through Table 3:

Table 3 - Technical description of the Project Activity²

Turbine Brand	Alstom
Turbine Type	Vertical Pelton
Turbine Serial Numbers	2073/ 2074
Number of units	2
Turbine	Rotations: 750 rpm
Water flow rate	4.75m ³ /s
Rated power	31.534 MW

Please see the technical details of the generators through Table 4:³

Generator Brand	Alstom
Generator Type	SAV 217/120/8
Generator Serial Numbers	G0N122, G0N123
Number of units	2
Rating	34500 kVA
Frequency	50 Hz

1.2 Sectoral Scope and Project Type

According to UNFCCC sectoral scopes definition for CDM projects, the Project Activity is included in the Sectoral Scope 1, category “Energy industries (renewable - / nonrenewable sources)”. The project is a single green field investment and is not part of a project group or bundle.

² The photographic evidence of turbine name plates is available to the DOE.

³ The photographic evidence of generator name plates is available to the DOE.

1.3 Project Proponent

Organization name	KARADENİZ HES ELEKTRİK A.Ş.
Contact person	Orçun Altınbaş
Title	ARGE Engineer
Address	Fahrettin Kerim Gökay cad. No:36 Altunizade 34662 Üsküdar – İstanbul - Turkey
Telephone	0090-216-5442400
Email	orcun.altinbas@iltekenerji.com.tr

1.4 Other Entities Involved in the Project

Organization name	Sekans Danışmanlık
Role in the Project	Consultant
Contact person	Sıla Duran
Title	Consultant
Address	
Telephone	-
Email	sila@sekansdanismanlik.com

1.5 Project Start Date

The project start date is 27/05/2010⁴ as it is the date of commissioning date of the project.

1.6 Project Crediting Period

A two times renewable crediting period of 10 years 0 month shall apply. The Project's total crediting period is: 27/05/2010 – 26/05/2020 (both days are included).

⁴ The Ministry Acceptance Protocols are available to the DOE.

The project is located in the Black Sea region of Turkey, within the province of Rize. The project activity is located on the Uzundere creek. The altitude at the powerhouse location is 817.59 m. The project site was treasury land. The coordinates of the Project Activity are 40°50'04.98"N and 40°45'03.29"E.⁵

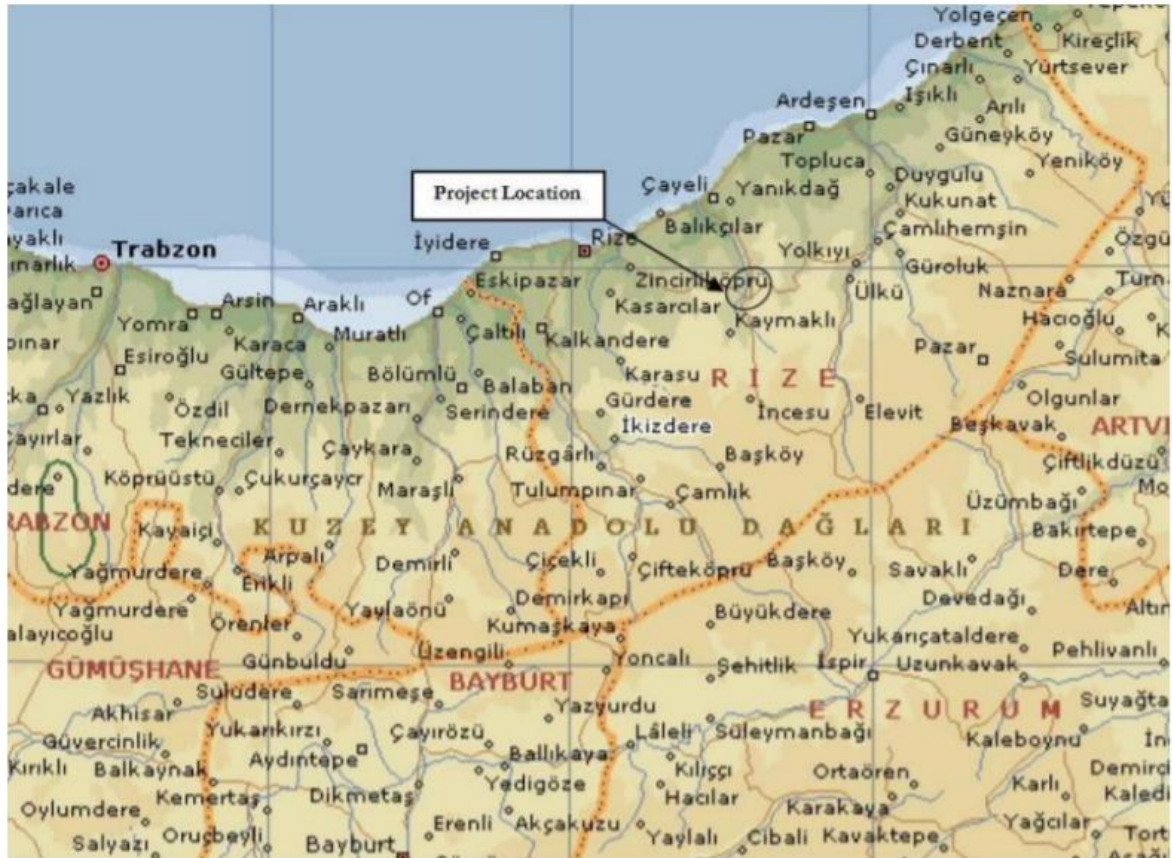


Figure 1. Location of the The Uzundere I 63.0 MW Hydroelectric Power Plant Project in Turkey Map

1.8 Title and Reference of Methodology

⁵ Google Earth screenshot of the Uzundere HPP has been provided to the DOE.

The emission reductions of the Project were calculated in accordance with the approved consolidated baseline and monitoring methodology ACM0002, Version 11: "Consolidated baseline methodology for grid-connected electricity generation from renewable sources".

Also, to prove additionality and to calculate the grid emission factor,

"Tool for the demonstration and assessment of additionality, Version 05.2 and

"Tool to calculate the emission factor for an electricity system", Version 02 are used.

1.9 Participation under other GHG Programs

The project has never been included in an emissions trading program or any other binding limits. It has neither received any kind of environmental credits nor been registered under any other GHG programs.

1.10 Other Forms of Credit

The project has not created another form of credit.

1.11 Sustainable Development

The project produces electricity from renewable energy sources using hydro as the power source and to contribute to Turkey's growing electricity demand through a sustainable and low carbon technology. The project displaces the same amount of electricity generated by the grid dominated with fossil fired power plants. The project generates 156,205 MWh annually. The project contributes to the Sustainable Development Goal, Affordable and Clean Energy. During this monitoring period, the project has generated 811,962.21 MWh as contributing to SDG 7 Affordable and Clean Energy.

The annual emission reduction estimated by the project is 87,318 tonnes CO₂eq, approximately. While this amount of emissions is mitigated, technology transfer is also realized as benefitting from hydropower. During this monitoring period, the project has contributed to SDG 13 Climate Action as reducing 453,882 tonnes CO₂.

The project contributes to improve the environmental situation in the region and in the country as avoiding fossil fuel-based electricity enhanced the air quality and help to reduce the negative effects on the climate. Through renewable technologies and hydro-based electricity sustainable and climate friendly development is promoted. The project contributes to the Sustainable Development Goal, Climate Action.

During construction and operational period, the project has created employment opportunities for the local community. As contributing to SDG 8 Decent Work and Economic Growth, 16 employees are working for the project activity.

The project contributes the economic development of the region by providing sustainable energy resources. The project provides workers with a safe and healthy work environment and is not complicit in exposing workers to unsafe or unhealthy work environments. Thus, the project contributes to the Sustainable Development Goal, Decent Work and Economic Growth.

2 SAFEGUARDS

2.1 No Net Harm

Within the scope of the project, all precautions have been taken for the environment during the design phase and the project will be implemented in line with the environmental law and related regulations.

The purpose of the project activity is to generate power in an efficient, clean, reliable and sustainable way with maximum respect on social and environmental aspects and to reduce emissions by partially substituting the electricity supply of fossil fuel fired power plants in Turkey. The consequences of the Project Activity with regards to economic, social and environmental impacts have been assessed during EIA process.

Regarding the wastes, hazardous wastes are handled appropriately in closed containers and transported by licensed transporters to the licensed processing and disposal facilities⁶.

Wastewater is collected through within the septic tank and is transferred through these wage truck⁷.

The domestic wastes are taken to the containers of the Municipality by the site personnel and the waste are transferred by the Municipality⁸.

There is no agricultural area in the project area and the lands were owned by the state.

There is not any negative environmental or socio-economic impact.

⁶ Hazardous waste transfer records are available to the DOE.

⁷ The waste water transfer receipts are available to the DOE.

⁸ The records are available to the DOE.

2.2 Local Stakeholder Consultation

During planning and construction phases, the project owner managed to conduct various unofficial and unstructured stakeholder participation meetings. In these meetings authorized people from KARADENİZ HES ELETRİK A.Ş. introduced project to the local people in terms of the size of the project, expected length of the construction period and to give details about how this project impacted their lives.

The feedback from the stakeholders was reflected on the project design and implementation. Also, a contact person from KARADENİZ HES ELEKTRİK A.Ş.. and relevant contact information was announced in the relevant meetings for ongoing communication.

During the construction phase of the project, nearly 120 workers were employed periodically. The local workers have been given priority during the hiring for plant operation. Also, during the construction, construction machines and equipment have been hired from the vicinity of the project site in order to contribute to the local economy of the region.

The Project has resulted in the creation of new jobs in the project region and improvement in local roads, contributing to living standards in the region. The project owner helped local people and the institutions in many ways, from renewing the electricity infrastructure to building roads in the forest to help forest workers in fighting against possible fires.

There was no specific environmental problem that concerned the locals more than others, and environmental issues need to be solved by the government. All people considered clear signs of climate change in the region in recent years.

The contact information of the plant responsible exist at the Mukhtars of Çataldere and İncesu villages, the project owner and local community are always in touch. The project owner regularly checks with the Mukhtar if any complaint or a request exists. Any complaint or need from the local community could directly be received by the project owner and appropriate contributions or improvements are made to the local community. Regarding the monitoring period, signed letters by the Mukhtars of Çataldere and İncesu (dated on 26/02/2021) have been provided as declaring that the related information has been available to the villagers and they did not have any complaint.

The project is operational and there is no update to the project design.

2.3 AFOLU-Specific Safeguards

N/A

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The description of the implementation and operational status of the project and the installed technology(ies), technical process and equipment are detailed in Section 1.1.

Other entity involved in the Project was GAIA Finansal Danismanlik, Turkey in the project description. However, it's been changed as Sekans Danışmanlık after the first monitoring period.

There hasn't been any event that may impact the GHG emission reductions or removals and monitoring during this monitoring period,

3.2 Deviations

3.2.1 Methodology Deviations

There are no methodology deviations applied during this monitoring period.

3.2.2 Project Description Deviations

The description of the implementation and operational status of the project and the installed technology(ies), technical process and equipment are detailed in Section 1.1. The project Proponent name has been changed as Karadeniz HES Elektrik A.Ş. after the first monitoring period⁹. Other entity involved in the Project was GAIA Finansal Danismanlik, Turkey in the project description. However, it's been changed as Sekans Danismanlik after the first monitoring period.

3.3 Grouped Projects

This is not a grouped project.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CMY}
Data unit	tCO ₂ e/MWh
Description	Baseline emission factor

⁹ The screenshot of the project license from the website of EMRA is available to the VVB.

Source of data	Registered PDD for the Project
Value applied	0.559
Purpose of data	Used for baseline emissions
Comments	-

Data / Parameter	$FC_{i,y}$
Data unit	ton or 1000 m ³
Description	Amount of fossil fuel type i consumed in the project electricity system
Source of data	As per the "Tool to calculate the emission factor for an electricity system"
Value applied	Please refer to Annex 3 in the PDD
Purpose of data	Data Used for emission reduction calculation
Comments	-

Data / Parameter	$NCV_{i,y}$
Data unit	TJ/kt or TJ/m ³
Description	Net calorific value of fossil fuel type i in year y
Source of data	As per the "Tool to calculate the emission factor for an electricity system"
Value applied	Please refer to Annex 3 in the PDD
Purpose of data	Data used to calculate the combined margin emission factor
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	$EG_{Facility,y}$
Data unit	MWh
Description	Quantity of net electricity delivered to the grid
Source of data	The data from the Electricity Meters are the basis for the settlement notification of EPIAS ¹⁰ . Data are gathered

¹⁰ PMUM has been replaced by EPIAS as of 01/09/2015 in Turkey. Retrospective data is accessible via EPIAS

	electronically from the meters by TEIAS and stored in secured website of EPIAS, which is accessible to project developer with a private password. For monitoring, EPIAS screenshots shall be used as source of data.
Description of measurement methods and procedures to be applied	These devices measure the net electricity supply to the national grid by the Project Activity, all losses before this point are on account of the project participant. Both metering instruments, which continuously monitor and measure the net electricity delivered by the Project Activity, are sealed and only accessible by TEIAS personnel. Official TEIAS data from the EPIAS web site are being used for calculating $EG_{Facility,y}$.
Frequency of monitoring/recording	Continuously monitoring and monthly readings
Value monitored	4,424.73 MWh for 2011 117,996.25 MWh for 2012 119,471.12 MWh for 2013 134,233.91 MWh for 2014 128,189.27 MWh for 2015 174,069.32 MWh for 2016 133,577.61 MWh for 2017

Monitoring equipment	Both exported and imported electricity are measured continuously by a power meter at the grid interface and recorded monthly.		
	Power meters:		
		Main Meter	Spare Meter
	Name	Ana Sayaç	Yedek Sayaç
	Brand	EMH	EMH
	Serial Number	9674605	9674606
	Latest Test Date of the Meters	15.10.2020	15.10.2020
	Accuracy Class	C-1SS	C-1S
	Power meters that were changed on 15.10.2020:		
		Main Meter	Spare Meter
Name	Ana Sayaç	Yedek Sayaç	
Brand	ACTARIS	ACTARIS	
Serial Number	53064430	53064429	
Accuracy Class	0.2 S	0.2 S	
Date of the First Index Protocol is 08/02/2010. ¹¹			
Latest two meter test dates are 06/11/2018 and 15/10/2020.			
QA/QC procedures to be applied	- A spare meter is used for crosschecking the accuracy and both meters are calibrated if required. - EPIAS records. are considered as the main source for the net electricity and the values are crosschecked with the Meter Reading Forms - TEIAS is responsible for calibration and maintenance of the devices. The periodic calibration or maintenance is under the responsibility of TEIAS and has been fixed		

¹¹ Protocol has been provided to the DOE.

	as once in 10 years ¹² . Since TEIAS meters are sealed by TEIAS, the project proponent cannot intervene with the devices. The periodic tests are executed on annual basis.
Purpose of data	Calculation of net electricity supplied to the grid and thus baseline emissions
Calculation method	Direct continuous measurement
Comments	-

Data / Parameter	CAP _{PJ}
Data unit	MW
Description	Installed capacity of the hydro power plant after the implementation of the project activity
Source of data	Project site
Description of measurement methods and procedures to be applied	Supplier information on the equipment
Frequency of monitoring/recording	Yearly
Value monitored	62.152 MWe (63.068 MWm)
Monitoring equipment	The data is monitored from the electricity generation license which was granted by Energy Market Regulatory Authority. The most up-to-date generation license has been provided. The installed capacity figure is 31.534 MWm / 31.076 MWe for each of the two turbines. The total installed capacity of the project is 63.068 MWm / 62.152 MWe.
QA/QC procedures to be applied	Supplier information on the equipment and existence of the equipment are being checked.
Purpose of data	Used for calculation of project emissions
Calculation method	-
Comments	-

¹² <http://www.mevzuat.gov.tr/Metin.Aspx?MevzuatKod=7.5.6381&MevzuatIliski=0&sourceXmlSearch>

Data / Parameter	APJ
Data unit	m ²
Description	Surface of the regulation pond after the implementation of the project activity, when the reservoir is full
Source of data	Project site
Description of measurement methods and procedures to be applied	The area was calculated by making coordinate readings around the border of the reservoir through topographical surveys or GPS readings. The technical drawing of the regulator area is based on topographical surveys. Based on the technical drawing, the reservoir area is calculated by a computer program.
Frequency of monitoring/recording	Yearly
Value monitored	7,275.54 ¹³
Monitoring equipment	-
QA/QC procedures to be applied	The technical drawing of the regulator area obtained from final design of the project is computerized and the potential lake area for the flood discharge (possible maximum water level) is calculated.
Purpose of data	Used for calculation of project emissions
Calculation method	-
Comments	-

4.3 Monitoring Plan

The objective of the monitoring plan is to ensure the complete, consistent, clear, and accurate monitoring and calculation of the emission reductions during the whole crediting period. The Project proponent is responsible for the implementation of the monitoring plan.

The Project Proponent is responsible for the overall management of the monitoring procedures including recording, data collection, calculating emission reductions and project emissions.

¹³ Please see the registered PD for the calculation. Calculation is also available in the ER Calculations file.

Please see below the management structure for the plant operation:

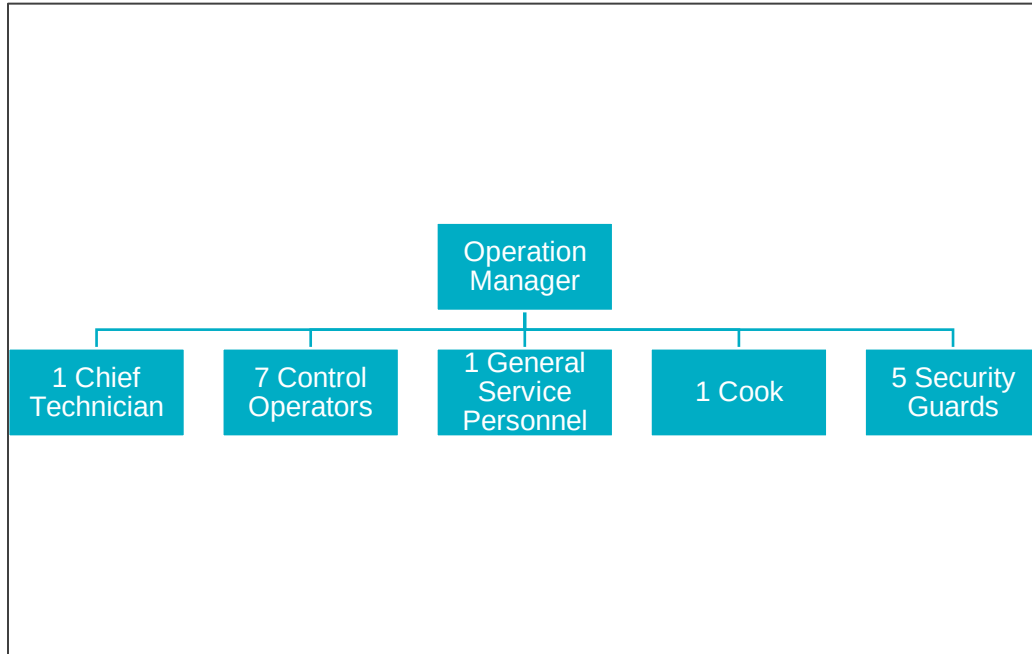


Figure 2. The management structure for the plant operation

The Operation Manager is responsible for the whole management of the Project Activity and the chief technician and control operators are responsible for the implementation of the Project Activity. In addition to the operators, there is one general service personnel, one cook and five security guards. In total, sixteen employees are working for the Project Activity.

Two power meters are installed at the grid interface of the project. One is the main meter and the other is spare meter of the main meter for cross-checking. Both meters are jointly inspected and sealed in order to be protected from interference by any of the parties. Both the main and secondary meters are owned and installed by the grid operator (TEIAS). The grid operator's is the only one authorized to deal with fixing, calibrating, or changing the meters, which is being done either by the grid operator or by a company authorized by the grid operator. During this monitoring period, there were no records of meter failure. In case of any urgent case TEIAS contacts the Plant Manager. Since the meters are within TEIAS' province, TEIAS executes all the procedures for handling non-conformities. Therefore, the Project Participant does not have any internal auditing for this purpose.

TEIAS is performing remote reading of the meters and monthly power meter readings are the basis for monitoring net electricity fed into the grid. A measuring protocol is prepared including day, peak and night hour electricity generation by the project owner and approved by governmental officers at the end of each month.

The primary source is the EPIAS records. Net electricity exported is crosschecked with the Meter Reading Forms. EPIAS is the financial settlement centre of TEIAS¹⁴. The Meter Reading Forms are filled by the project owner and approved by the governmental officers. Additionally, the remote reading by the governmental body is also available. The website of EPIAS is accessible to Project Proponent with their unique user ID and password. Once accessed, the Project Proponent is able to call electricity generation and consumption reports of their own projects. The same reports are used by the Project Proponent for invoicing TEIAS. The electricity generation data is reported monthly basis.

Quality Control and Assurance (QC/QA)

Both the main and spare (control) meters are owned and installed by the grid company. Both meters are jointly inspected and sealed in order to be protected from interference by any of the parties.

The grid operator's Metering Officer should be notified of any failure of one of the meters. He is the only one authorized to deal with fixing, calibrating, or changing the meters, which is being done either by the grid operator or by a company authorized by the grid operator. During this monitoring period, there were no records of meter failure. The data is hourly recorded by the personnel on the plant. TEIAS Load Dispatching Center also executes remote reading during 24-hour period. The data could be remotely accessed by TEIAS.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The baseline emission BE_y (tCO₂e) during the monitoring period results from:

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$$

Where:

BE_y

Baseline emissions in year y (tCO₂e/y)

¹⁴ PMUM has been replaced by EPIAS as of 01/09/2015 in Turkey during the verification period. Retrospective data is accessible via EPIAS.

$EG_{PJ,y}$ Quantity of net electricity supplied to the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,CM,y}$ Combined margin CO₂ emission factor for grid connected power generation in year y calculated by using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

$EF_{grid,CM,y}$ value is fixed as 0.559 tCO₂/MWh for during the crediting period. Table 2 shows the baseline emissions for the verification period:

Table 3 - Baseline emissions

Month	(A) Electricity supplied to the grid [MWh]	(B) Electricity consumpt ion from the grid [MWh]	(C) = (A) - (B) EG (ID 8) Net electricity supplied to the grid [MWh]	EF [tCO ₂ /MWh]	Baseline emission: ER = EG * EF [t CO ₂ -eq]
Dec-11	4,466.60	41.87	4,424.73	0.559	2,473
Jan-12	3,512.73	45.21	3,467.52	0.559	1,938
Feb-12	2,626.52	51.56	2,574.96	0.559	1,439
Mar-12	2,631.20	50.36	2,580.84	0.559	1,443
Apr-12	15,738.76	18.03	15,720.73	0.559	8,788
May-12	34,574.15	7.50	34,566.65	0.559	19,323
Jun-12	26,353.81	10.62	26,343.19	0.559	14,726
Jul-12	12,646.60	22.40	12,624.20	0.559	7,057
Aug-12	7,664.70	25.72	7,638.98	0.559	4,270
Sep-12	1,444.42	31.37	1,413.05	0.559	790
Oct-12	1,647.04	35.79	1,611.25	0.559	901
Nov-12	5,740.94	28.95	5,711.99	0.559	3,193
Dec-12	3,776.22	33.33	3,742.89	0.559	2,092
Jan-13	2,888.62	39.72	2,848.90	0.559	1,593
Feb-13	2,824.39	35.20	2,789.19	0.559	1,559
Mar-13	6,349.87	39.19	6,310.68	0.559	3,528
Apr-13	12,407.59	20.02	12,387.57	0.559	6,925
May-13	25,453.14	10.48	25,442.66	0.559	14,222
Jun-13	21,744.67	13.81	21,730.86	0.559	12,148
Jul-13	11,658.69	22.08	11,636.61	0.559	6,505

Aug-13	6,595.68	25.95	6,569.73	0.559	3,672
Sep-13	9,114.10	21.98	9,092.12	0.559	5,082
Oct-13	10,653.70	24.16	10,629.54	0.559	5,942
Nov-13	5,826.30	29.45	5,796.85	0.559	3,240
Dec-13	4,275.72	39.31	4,236.41	0.559	2,368
Jan-14	3,361.44	39.64	3,321.80	0.559	1,857
Feb-14	2,968.84	37.80	2,931.04	0.559	1,638
Mar-14	6,005.17	33.29	5,971.88	0.559	3,338
Apr-14	16,458.56	20.84	16,437.72	0.559	9,189
May-14	34,230.09	7.71	34,222.38	0.559	19,130
Jun-14	22,507.91	11.82	22,496.09	0.559	12,575
Jul-14	10,583.06	18.28	10,564.78	0.559	5,906
Aug-14	6,458.15	24.74	6,433.41	0.559	3,596
Sep-14	7,100.02	26.33	7,073.69	0.559	3,954
Oct-14	10,302.19	22.08	10,280.11	0.559	5,747
Nov-14	7,575.70	25.46	7,550.24	0.559	4,221
Dec-14	6,987.13	36.36	6,950.77	0.559	3,885
Jan-15	4,176.96	38.07	4,138.89	0.559	2,314
Feb-15	3,321.44	36.49	3,284.95	0.559	1,836
Mar-15	4,986.26	37.83	4,948.43	0.559	2,766
Apr-15	10,915.62	26.76	10,888.86	0.559	6,087
May-15	32,658.76	9.29	32,649.47	0.559	18,251
Jun-15	33,384.71	7.08	33,377.63	0.559	18,658
Jul-15	12,896.47	17.29	12,879.18	0.559	7,199
Aug-15	7,547.76	24.41	7,523.35	0.559	4,206
Sep-15	5,050.95	26.27	5,024.68	0.559	2,809
Oct-15	10,187.36	21.75	10,165.61	0.559	5,683
Nov-15	3,016.20	8.87	3,007.33	0.559	1,681
Dec-15	301.67	0.78	300.89	0.559	168
Jan-16	4,768.32	35.33	4,732.99	0.559	2,646
Feb-16	4,952.38	33.43	4,918.95	0.559	2,750
Mar-16	7,217.78	34.80	7,182.98	0.559	4,015
Apr-16	20,643.93	15.37	20,628.56	0.559	11,531
May-16	31,782.44	1.80	31,780.64	0.559	17,765
Jun-16	32,305.15	1.04	32,304.11	0.559	18,058
Jul-16	25,485.91	7.39	25,478.52	0.559	14,242
Aug-16	9,804.04	18.30	9,785.74	0.559	5,470
Sep-16	10,623.82	12.92	10,610.90	0.559	5,931
Oct-16	12,932.07	20.86	12,911.21	0.559	7,217
Nov-16	8,695.79	12.02	8,683.77	0.559	4,854
Dec-16	5,082.83	31.88	5,050.95	0.559	2,823

Jan-17	4,233.09	34.45	4,198.64	0.559	2,347
Feb-17	2,981.32	38.45	2,942.87	0.559	1,645
Mar-17	4,697.28	37.56	4,659.72	0.559	2,605
Apr-17	12,418.53	12.33	12,406.20	0.559	6,935
May-17	34,851.18	0.54	34,850.64	0.559	19,482
Jun-17	32,923.23	1.03	32,922.20	0.559	18,404
Jul-17	12,840.76	9.92	12,830.84	0.559	7,172
Aug-17	5,974.25	23.45	5,950.80	0.559	3,326
Sep-17	5,560.28	21.54	5,538.74	0.559	3,096
Oct-17	10,831.07	20.52	10,810.55	0.559	6,043
Nov-17	6,494.46	28.05	6,466.41	0.559	3,615
2011 Vintage (02.12.2011- 31.12.2011)	4,466.60	41.87	4,424.73	0.559	2,473
2012 Vintage (01.01.2012- 31.12.2012)	118,357.09	360.84	117,996.25	0.559	65,959
2013 Vintage (01.01.2013- 31.12.2013)	119,792.47	321.35	119,471.12	0.559	66,784
2014 Vintage (01.01.2014- 31.12.2014)	134,538.26	304.35	134,233.91	0.559	75,036
2015 Vintage (01.01.2015- 31.12.2015)	128,444.16	254.89	128,189.27	0.559	71,657
2016 Vintage (01.01.2016- 31.12.2016)	174,294.46	225.14	174,069.32	0.559	97,304
2017 Vintage (01.01.2017- 30.11.2017)	133,805.45	227.84	133,577.61	0.559	74,669
Total	813,698.49	1,736.28	811,962.21	0.559	453,882

5.2 Project Emissions

There is no project emission resulting from the reservoir area of the Project Activity as the power density of the project is greater than 10W/m². Accordingly, The Project Activity's power density was calculated as below:

$$=62,152,000 \text{ W}/7,275.54 \text{ m}^2^{15}$$

$$= 8,542.6 \text{ W}/\text{m}^2$$

Therefore, $PE = 0$

5.3 Leakage

No leakage is to be accounted by the Project Activity. This is in line with the registered PD and applicable methodology ACM0002, Version 11. Therefore, the leakage from the Project Activity is zero.

5.4 Net GHG Emission Reductions and Removals

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y Emission reductions in year y (t CO₂e/yr)

BE_y Baseline emissions in year y (tCO₂e/y)

PE_y Project emissions in year y (t CO₂e/yr)

The baseline emission BE_y (tCO₂e) during the monitoring period results from:

$$BE_y = EG_{PJ, y} * EF_{\text{grid, CM, y}}^{16}$$

Where:

¹⁵ Please see the registered PD for the calculation. Calculation is also available in the ER Calculations file.

¹⁶ Please see ER calculations_Uzundere HPP.

BE_y	Baseline emissions in year y (tCO ₂ e/y)
$EG_{PJ,y}$	Quantity of net electricity supplied to the grid as a result of the implementation of the CDM project activity in year y (MWh)
$EF_{grid,CM,y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated by using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO ₂ /MWh)

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2011	2,473	0	0	2,473
2012	65,959	0	0	65,959
2013	66,784	0	0	66,784
2014	75,036	0	0	75,036
2015	71,657	0	0	71,657
2016	97,304	0	0	97,304
2017	74,669	0	0	74,669
Total	453,882	0	0	453,882

Project emissions apply only for geothermal power plants, solar thermal power plants and for some hydro power plants.

Therefore, $PE_y = 0$

In total, for this monitoring period:

$$ER_y = BE_y - PE_y$$

$$=453,882-0$$

$$=453,882$$